



ScholarNest

Spark Azure Databricks

Databricks Spark Certification and beyond

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Azure
Databricks



Absolute Beginner to Specialization in Apache Spark and Azure Databricks



Spark Data Types

Every data processing system defines a set of data types.
If you have worked with relational databases, you know it already.
A database defines a list of data types, and you can use those types
to define table schema.
Similarly, Spark also comes with a list of types.

Go to the latest Pyspark documentation page, and navigate to the Data Types page via the API reference option. (Reference :

<https://spark.apache.org/docs/latest/api/python/reference/pyspark.sql.html#data-types>)

You will see a list of Spark data types.

Spark SQL — PySpark 3.2.1 docu

spark.apache.org/docs/latest/api/python/reference/pyspark.sql.html#data-types

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- pyspark.sql PandasCogroupedOps
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- pyspark.sql SparkSession.builder
- pyspark.sql SparkSession.builder.config
- pyspark.sql SparkSession.builder.getOrCreate
- pyspark.sql SparkSession.builder.newSession

Data Types

<code>ArrayType(elementType[, containsNull])</code>	Array data type.
<code>BinaryType</code>	Binary (byte array) data type.
<code>BooleanType</code>	Boolean data type.
<code>ByteType</code>	Byte data type, i.e.
<code>DataType</code>	Base class for data types.
<code>DateType</code>	Date (datetime.date) data type.
<code>DecimalType([precision, scale])</code>	Decimal (decimal.Decimal) data type.
<code>DoubleType</code>	Double data type, representing double precision floats.
<code>FloatType</code>	Float data type, representing single precision floats.
<code>IntegerType</code>	Int data type, i.e.

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Here is a summarized list of all the Spark types.

S.No.	Spark Types	Spark SQL Types	Description
1.	DataType	NA	Base class for all Spark types
2.	ByteType	BYTE, TINYINT	8 bit signed integer between -128 to 127
3.	ShortType	SHORT, SMALLINT	16 bit signed integer between -32768 to 32767
4.	IntegerType	INT, INTEGER	32 bit signed integer between -2147483648 to 2147483647
5.	LongType	LONG, BIGINT	64 bit signed integer between 9223372036854775808 to 9223372036854775807
6.	FloatType	FLOAT, REAL	32 bit single precision floating point number
7.	DoubleType	DOUBLE	64 bit double precision floating point number
8.	DecimalType	DECIMAL, DEC, NUMERIC	Maximum precision
9.	BooleanType	BOOLEAN	
10.	StringType	STRING	
11.	DateType	DATE	
12.	TimestampType	TIMESTAMP	
13.	BinaryType	BINARY	
14.	ArrayType	ARRAY	
15.	MapType	MAP	
16.	StructType	STRUCT	
17.	StructField		
18.	NullType		
19.	YearMonthIntervalType	INTERVAL YEAR, INTERVAL YEAR TO MONTH, INTERVAL MONTH	
20.	DayTimeIntervalType	INTERVAL DAY, INTERVAL DAY TO HOUR, INTERVAL DAY TO MINUTE, INTERVAL DAY TO SECOND, INTERVAL HOUR, INTERVAL HOUR TO MINUTE, INTERVAL HOUR TO SECOND,	

So we have twenty types. The first four types are usually not used much in PySpark. The first one is the DataType class. This guy is the base class of all the Spark Data types. You can ignore it because you are not likely to use it anywhere. The other three are ByteType, ShortType, and the IntegerType. Spark defined them as 8-bit, 16-bit, and 32-bit signed integers. However, we do not recommend using these types. The best practice is to use the LongType for integers.

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16.	StructType	STRUCT	
17.	StructField		
18.	NullType		
19.	YearMonthIntervalType	INTERVAL YEAR, INTERVAL YEAR TO MONTH, INTERVAL MONTH	
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The Byte, Short, and Integer are designed in a data system to lower the memory footprint of the data. Spark performs in-memory computations. So using right-sized data types might help to use less memory. But saving memory over data types is not significant for real-life scenarios.

The IntegerType is 32-bit, and that guy is reasonably large. It can hold numbers a little larger than two billion. So we can use the IntegerType if we know we are not going to go above a couple of billion.

The long goes over nine quintillions, and that guy is relatively safe. You might not even reach quintillions. So one reason for recommending LongType and ignoring other integer types is to avoid the risk of crossing the limits.

ByteType valid values: -128 to +127

ShortType valid values: -32768 to 32767

- Above ranges are too small
- Risk of going out of range and cause runtime exception
- Spark implicitly converts out of range values to null

IntegerType valid values: -2147483648 to 2147483647

LongType valid values: - 9223372036854775808 to
9223372036854775807

- Int is safe to use if you are sure of staying below 2 billion
- Long is recommended

Now, look at the savings.

IntegerType is 4 bytes, and LongType is 8 bytes. So for each column, you can save 4 bytes of memory. Let's assume you have 1 million records and one IntegerType column. You could have taken LongType, but you took IntegerType to save 4 bytes per record. You can see the saving in the table below.

Is it worth trying to save memory by choosing IntegerType instead of LongType if you do not have billions of records? And you keep on increasing the risk of crossing the limits.

So, LongType is recommended for integers unless you are too sure about the limits.

IntegerType: 4 Bytes
LongType: 8 Bytes

Long Size	Int Size	Net Savings	No of Columns	Number of Records	Total Savings	Savings in MB
8	4	4	1	1	4	0.000004
8	4	4	1	1000000	4000000	4
8	4	4	10	1000000	40000000	40
8	4	4	10	10000000	400000000	400
8	4	4	1	2000000000	8000000000	8000

The following three types are for decimal numbers. The float type is a 32-bit single-precision data type, and the double is 64-bit double precision. Most languages recommend using 64-bit double, and Spark is no different. We recommend using double for most of your floating-point numbers.

You can use DecimalType if you need maximum precision. However, most of your requirements are met by the double, and that one is recommended. So we have four integer types and three decimal types in Spark. However, you will be primarily using LongType and DoubleType in your application. These guys reduce the risk and fit most of your requirements.

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9.	BooleanType	BOOLEAN	
10.	StringType	STRING	
11.	DateType	DATE	
12.	TimestampType	TIMESTAMP	
13.	BinaryType	BINARY	
14.	ArrayType	ARRAY	
15.	MapType	MAP	
16.	StructType	STRUCT	

The following five data types are straightforward. The BooleanType is for true/false boolean values. StringType is to store string values. The Datetype stores date values, and TimestampType is for a date with a time component. And the BinamyType is for raw bytes.

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11.	DateType	DATE	
12.	TimestampType	TIMESTAMP	
13.	BinaryType	BINARY	
14.	ArrayType	ARRAY	
15.	MapType	MAP	
16.	StructType	STRUCT	

Then we have three complex data types: ArrayType, MapType and StructType.

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11.	DateType	DATE	
12.	TimestampType	TIMESTAMP	
13.	BinaryType	BINARY	
14.	ArrayType	ARRAY	
15.	MapType	MAP	
16.	StructType	STRUCT	

This sheet also shows the SQL keywords for these data types. For example, we use LongType in PySpark, and we can use LONG or BIGINT in Spark SQL. The LONG and BIGINT are the same in Spark SQL, and they internally represent LongType in Spark.

Similarly, you can use StringType in PySpark and STRING in Spark SQL. They mean the same thing.

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13.	BinaryType	BINARY	
14.	ArrayType	ARRAY	
15.	MapType	MAP	
16.	StructType	STRUCT	

We are using Spark SQL data types to define the table schema. That's how you can use the Spark SQL data types.

```
create table if not exists demo_db.fire_service_calls_tbl(  
  CallNumber integer,  
  UnitID string,  
  IncidentNumber integer,  
  CallType string,  
  CallDate string,  
  WatchDate string,  
  CallFinalDisposition string,  
  AvailableDtTm string,  
  Address string,  
  City string,  
  Zipcode integer,  
  Battalion string,  
  StationArea string,  
  Box string,  
  OriginalPriority string,  
  Priority string,  
  FinalPriority integer,  
  ALSUnit boolean,  
  CallTypeGroup string,  
  NumAlarms integer,  
  UnitType string,  
  UnitSequenceInCallDispatch integer,  
  FirePreventionDistrict string,  
  SupervisorDistrict string,  
  Neighborhood string,  
  Location string,  
  RowID string,  
  Delay float  
) using parquet
```

Here is how we use Dataframe Schema in PySpark. We learned two approaches. The first one shown above uses using DDL schema definition, and the second one shown below uses PySpark syntax. And this is where we use Spark data types.

```
flight_schema_ddl = """FL_DATE DATE, OP_CARRIER STRING, OP_CARRIER_FL_NUM INT, ORIGIN STRING,  
                        ORIGIN_CITY_NAME STRING, DEST STRING, DEST_CITY_NAME STRING, CRS_DEP_TIME INT, DEP_TIME INT,  
                        WHEELS_ON INT, TAXI_IN INT, CRS_ARR_TIME INT, ARR_TIME INT, CANCELLED STRING, DISTANCE INT"""
```

```
flight_schema = StructType([  
    StructField("FL_DATE", DateType()),  
    StructField("OP_CARRIER", StringType()),  
    StructField("OP_CARRIER_FL_NUM", IntegerType()),  
    StructField("ORIGIN", StringType()),  
    StructField("ORIGIN_CITY_NAME", StringType()),  
    StructField("DEST", StringType()),  
    StructField("DEST_CITY_NAME", StringType()),  
    StructField("CRS_DEP_TIME", IntegerType()),  
    StructField("DEP_TIME", IntegerType()),  
    StructField("WHEELS_ON", IntegerType()),  
    StructField("TAXI_IN", IntegerType()),  
    StructField("CRS_ARR_TIME", IntegerType()),  
    StructField("ARR_TIME", IntegerType()),  
    StructField("CANCELLED", IntegerType()),  
    StructField("DISTANCE", IntegerType())  
])
```

We have four more types in Spark listed below.

The StructField is not an actual data type. We use it to define a dataframe column in PySpark. We use this guy to define a schema. This class is defined in the Spark types package, but it is not an actual data type. It is a utility class.

The NullType represents no data type. You are not likely to use this class. Spark assigns a NullType to a column when it cannot infer the schema for a dataframe column. But we do not recommend using schema inference. So the NullType is not used anywhere if you are not using schema inference.

The remaining two types are to represent intervals. These types are not yet available to PySpark. These guys are only available in Scala, Java, and Spark SQL. PySpark doesn't have anything for these two guys. So you cannot explicitly use them in PySpark, but we can use them in Spark SQL. However, future versions might bring these to PySpark as well.

```
17.      StructField
18.      NullType
19.      YearMonthIntervalType      INTERVAL YEAR,
                                   INTERVAL YEAR TO MONTH,
                                   INTERVAL MONTH
20.      DayTimeIntervalType        INTERVAL DAY,
                                   INTERVAL DAY TO HOUR,
                                   INTERVAL DAY TO MINUTE,
                                   INTERVAL DAY TO SECOND,
                                   INTERVAL HOUR,
                                   INTERVAL HOUR TO MINUTE,
                                   INTERVAL HOUR TO SECOND,
```



Thank You
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